

Functions

Brainster Web Development Academy



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Functions

You need to perform some task many times. Let's say it is about 10 lines of code and you are doing it 10 times.

If we do the math, that's **10 x 10 = 100 lines of code**.

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What if you can make a "template-of-the-action", write it only once, and call it, use it how much you like?



Functions

Let's say we have 3 arrays numbers.

For each array we should find the highest number(maximum)

The usual method is something like:

1. Declare the arrays
2. Make for loop for each arr
3. Print the final result



Example without a function

1. You should rewrite **all the code** for every array
2. It is easier to make a mistakes while typing
3. It takes more memory, lines of code, variables ...
4. Imagine if you had to check for 100 examples ?!



Example with a function

1. You made a 'template'(function) once and use it **as much as you want**
2. The risk of **making mistakes** is smaller
3. If you wanted to change the condition of the function, you just change it once, not 3 times
4. Less code, less memory
5. Cleaner code



Declaring functions

```
function functionName(parameters) {  
  // Do stuff here  
}
```

```
let functionName = function(parameters) {  
  // Do stuff here  
}
```

1. **function** is a keyword that tells JavaScript you're defining a function.
2. **functionName** is the name of the function.
3. **parameters** is optional. It is a comma-separated list of variables you wish to declare for your function. They can be assigned values when you use the function.



Using (invoking) functions

Once you declared your function, you can use (or invoke, or call, or execute) it **by writing the name of the function, followed by parenthesis ()**;

You can use them **as many times as you like**.

Declaring a function:

```
function sayHello() {  
  console.log('Hello world!');  
}
```

Using/invoking a function:

```
sayHello();
```



Example - declaration

Let's say you wish to write a **function** called **sayName** that logs the **firstName** and **lastName** of a person.

The function looks like this:

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```
function sayName(firstName, lastName) {  
  console.log('firstName is ' + firstName);  
  console.log('lastName is ' + lastName);  
}
```



Example - invocation

John is the first name, **Smith** is the last name. To get the function to work correctly, I pass John, as the **first** argument, and Smith as the **second** argument:

```
sayName('John', 'Smith');
```

```
// firstName is John
```

```
// lastName is Smith
```



Example - without parameters

If you declared an parameter, but did not pass an argument to it, your parameter would be **undefined**.

```
sayName();
```

```
// firstName is undefined
```

```
// lastName is undefined
```



Example - default parameter

```
function sayName(firstName, lastName="Doe") {  
  console.log('firstName is ' + firstName); // John  
  console.log('lastName is ' + lastName); // Doe  
}
```

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```
sayName("John");
```



The return keyword

return specifies the value that a function sends back to the place from which it was called.

Example:

```
function add(a, b) {  
  return a + b;  
}
```

let sum = add(5, 3); // sum will be 8 because the function returns 5 + 3

If no value is specified or if the function does not have a return statement, the function returns **undefined**.



Exercise I

Write a function that:

1. Produces a prompt that asks for a color name (red, blue, purple...)
2. Use the color from the prompt the paint the page in the written color
3. Invoke the function at page load



Exercise II

1. **Produce a prompt asking for a number**
2. **Define a function that:**
 - a) Takes a number as an argument
 - b) Determines whether that number is even or odd
 - c) Writes in the document whether the number is even or odd
3. **Invoke the function using the number from the prompt at 1**



Exercise III

1. **Create an unordered list with some elements (any number)**
2. **Prompt for a number**
3. **Create a function that:**
 - a) Takes a number argument
 - b) Paints in red an element from the unordered list at the index of the number argument above
4. **Invoke the function using the number prompt at 1**



Homework

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1. **Prompt for a number**
2. **Define a function that:**
 - a) Generates a DOM element from 1 up to a given number argument
 - b) For any number divisible by 3, print “Fizz” as the innerHTML of the DOM element.
 - c) For any number divisible by 5, print “Buzz” as the innerHTML of the DOM element.
 - d) For any number divisible by both 3 and 5, print “FizzBuzz” as the innerHTML of the DOM element.
 - e) For any other number, just print the value of that number as the innerHTML of the DOM element.
3. **Invoke the function using the number from the prompt as 1.**

